

Use the pre-worksheet to collect information that may help with filling out the scorecards.

[DOI: 10.5281/zenodo.15122045](https://doi.org/10.5281/zenodo.15122045)

[Repository Crisis Scorecards Instructions](#) Date:

Descriptor	Question	Answer
Metadata data volume	What is the total volume of metadata and data?	
Bandwidth availability	How much bandwidth is available for upload and download?	
Irreplaceable datasets	Are there datasets that cannot be obtained elsewhere?	
Metadata schemas	How many and which metadata schemas can be outputted?	
Packaging	Are there any datasets with an archive copy, or an archive with an archive copy?	
Data types	Are there any datasets with an archive copy, or an archive with an archive copy?	
Static datasets	Are there any datasets with an archive copy, or an archive with an archive copy?	
Considerations	Are there any datasets with an archive copy, or an archive with an archive copy?	
Stored separately?	Are there any datasets with an archive copy, or an archive with an archive copy?	
Location	Are there any datasets with an archive copy, or an archive with an archive copy?	
Public Plans	Are there any datasets with an archive copy, or an archive with an archive copy?	
Service Level Agreements	Are there any datasets with an archive copy, or an archive with an archive copy?	

Repository Name:

Organization:

Project:

Resiliency Considerations
can be useful in making these plans.
resources in the event of a crisis.
help in prioritizing efforts and time.

[Ex: Frictionless Data Package](#)
multiple paragraphs or multiple scorecards for each project.

best data management & governance practices.

[List webpages or DOIs to plans.](#)

The Repository Resilience Scorecard is used to help digital data repository staff assess cri
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[Repository Crisis Scorecards Instructions](#)

Repository Descriptors			Resiliency Metric	
			No resiliency	
Big Picture Question	Descriptor	Descriptor definition	Class 1	Class 2
Section Theme: Single Repository Dataset Replication				
<i>This section is filled out from the viewpoint of the organization looking at its dataset</i>				
How many dataset copies does the repository have?	Total number of logical copies	Number of discrete logical copies ma	One copy on one physical device	Multiple copies on one physical device
	Total number of storage medium types	Digital storage medium physical devices includes: hard disk drives, solid state drives, tape drives, Blu-ray discs, cloud storage	One type of storage medium	Two types of storage medium
	Total number of offline copies	Dataset copies that are held offline in an environment controlled, secure location to prevent bit rot or tampering.	Zero offline copies	One offline copy
	Total number of national repository facilities storing copies	Facilities are managed by the repository and are geographically separated and grid separated, ex. Los Angeles, USA and Richmond, USA.	One data facility	Two data facilities, geographically separated
	Total number of extranational repository facilities storing copies	Facilities are managed by the repository and are geographically separated and grid separated, ex. Los Angeles, USA and Tokyo, Japan. The facilities could be self-owned or a commercial cloud service.	Zero international data facilities	One international data facility

Section Theme: Repository Infrastructure

Repository Staff and Training	Are there redundant staff and requisite training for the repository? Can staff cover for each other in case of unavailability?	Each staff member is specialized and unable to do each other's role, or there is only one staff member. There is one staff member in each role and staff in each role can cover for staff in other roles.
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Can the physical operations of the repository run smoothly when in a crisis?	Repository Supported Virtual Data Transfer	Do bandwidth limitations impede data transfer options for critical datasets from the repository?	Data volume is too large to effectively transfer all data within one month.	Data volume is too large to effectively transfer all data within one week.
	Repository Supported Physical Data Transfer	What is the capability to send datasets physically in an emergency, ex. hard drives in a van?	There are no plans or training to physically move datasets, and/or it would take more than 48 hours.	There is capability to break down and physically move datasets in two days notice.
	Physical Plant Infrastructure Capability	How stable are power, HVAC, plumbing, building maintenance? (Should this question be split further?)	There are no redundant systems at any level.	Some systems have redundancy, others don't.

Section Theme: Routine Operations

Are processes and documentation in place to ensure resiliency on a routine basis?	Backup Schedule	How often are backups performed of the repository?	No regular updates or backups, or no device to backup to.	Backups are manual, irregular, and on the same device, and/or backups are not tested regularly.
	Service Level Agreements	Are there Service Level Agreements (SLAs) in place to communicate to users when they can expect services to be resumed in case of problems?	No SLAs are defined.	SLAs are defined for some operations.
	Continued Training/Professional Development	Do staff have regular training opportunities?	No required training for repository work or time allocated for staff improvement.	Required training for staff but no time/funds for staff improvement via optional conferences, trainings, or certifications.

Section Theme: Software

Is the software currently being used to manage datasets integral to the accessibility	Software as Requirement for Interpreting Datasets	Is the software a critical piece needed to understand and interact with the data and metadata? Ex. NetCDF library and tools	The software is integral to using the data and metadata efficiently, and datasets would not be accessible without the software.	There are multiple pieces of software that can be used to interact with the data and metadata that are interchangeable.
	Software Packaging and Backup Format	How are the systems software delivered by the developers and backed up by the repository? Is it possible for the public to get the software?	The software is not compiled by the developer to be delivered as a binary executable, and may be multiple files spread across directories.	
	Software Availability and Licensing	Are the systems software open source and unencumbered by licenses? If proprietary, how easy is it to acquire a license?	The software is proprietary and it is hard or impossible to get a license to use it.	The software is open source but has been abandoned/no updates to it in a while, OR the software is proprietary but still being supported by a company.
	Ease of Software Deployment	Is the software well documented and easy to install? Are there images or virtual machines to ease deployment? Are there sane initial configurations to use the software?	There is very little code and supporting documentation. Source code is difficult to understand and manage or not available.	Source code is well documented and easy to install by a technical user. Configuration may take some time. There is very little support for non-technical users to deploy the software.
	Maintainability of Software	How easy is it to maintain and extend the code?	The software is written in a language(s) that are not as popular and/or leverage unmaintained or expensive libraries. Code discipline is weak and the codebase would be greatly improved by a rewrite.	The software is in a common language and has decent code discipline, documentation, and testing. The software may rely on some libraries with weak support, or need some code changes to be supported by more robust communities.

of the holdings? Could this software be replaced with other software easily when considering potential costs to use and maintain the software, and use and maintain the holdings? Is the software publicly accessible outside of the repository?

Ease of Use of Software

How easy is it for an experienced non-programming user to work with the software to curate and access datasets?

Curators and users require a long training period to use the software. Documentation is lacking or non-existent. Programmatic access is lacking or non-existent.

Curators or users might have trouble using the software. Documentation exists but could be filled out. Some defaults are provided when performing common actions.

Section Theme: Cybersecurity and Access Control

Logging and Change Monitoring

How are changes tracked or logged? Can changes be rolled-back? What processes exist to ensure unauthorized changes are noted, alerted, and corrected?

There is no automated logging activated at the OS or repository systems level, and manual logging of actions is ad hoc.

There is automated logging at the OS level. There may be automated logging at the repository systems level, but monitoring of logs are ad hoc and infrequent. Logs may not be comprehensive, understandable, or easily accessible.

Access Control and Deletion Privileges

Are permissions appropriately scoped for each account? Are policies in place to limit accidental or hostile access to critical systems? Are editing and deletion privileges restricted to authorized users and clearly documented?

Staff permissions are not differentiated. There are no clear policies to follow for deletion of deposits, and no logs or records of deletions exist.

There are some accounts with higher permissions allowed to delete data. Some policies may be defined for deletion of data, but no additional safeguards.

Are best practices in place to ensure cyber security?

Cybersecurity Staffing	Are there dedicated roles for cybersecurity, and how many are there in the repository and larger organization? Do cybersecurity personnel have appropriate qualifications?	There are no cybersecurity professionals on staff or available through a larger organization.	There are cybersecurity professionals available through a larger organization.
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Section Theme: Sustainability

Data or Systems Modernization or Migration Plans	Does the repository have plans for the modernization of non versionless (ex. RAW image format, text) datasets and systems to new formats, tools, audiences?	Datasets/data systems are frozen, no data modernization capacity.	Dataset/data systems modernization tools and/or tutorials are available, but new versions of datasets are not hosted.
Repository Shutdown Plans	Does the repository have a clear plan for systematic shutdown of systems that preserves the data?	No shutdown plan exists.	There are plans for a systematic shutdown of systems but not transfer of datasets.
Continuity Agreement (Intranational)	Does the repository have a documented agreement for relocation or transition of holdings to another intranational repository in the event that the repository can no longer fulfill its mandate due to funding or shifted priorities?	No documented plan or agreement exists.	There is a formal agreement but lacking detailed plans.
Continuity Agreement (Extranational)	Does the repository have a documented agreement for relocation or transition of holdings to another extranational repository in the event that the repository can no longer fulfill its mandate due to funding or shifted priorities?	No documented plan or agreement exists.	There is a formal agreement but lacking detailed plans.
Parent Institutional Continuity Plans	What happens to the repository if the larger institution is undergoing a crisis?	There are no plans for what would happen to the repository in case of larger institutional crisis.	

What agreements and plans are in place to preserve the mission of the repository in the face of crisis or after cessation of activities?

Funding Diversity

How many funding streams to the upkeep of a repository? If one or funding streams suddenly disappeared, would operations be heavily impacted?

There is one main funding stream and few or no smaller funding streams. Losing the main funding stream would shutdown the repository.

There are one or two main funding streams and a few smaller funding streams. Losing one of the main funding streams would heavily impact the repository.

Section Theme: Network

Professional Networks

Is the repository a member of any professional networks, national or extranational, that could participate in reciprocal data sharing agreements?

The repository is not a member of any professional networks.

The repository is a member of one or more professional networks, but there are no reciprocal data sharing agreements.

Sibling Repositories

Is the repository a member of a larger organization with access to libraries or archives that could support dataset deposits?

There are no similar libraries, archives, or repositories in the larger organization.

There are other libraries, archives, or repositories in the larger organization but there are no explicit agreements on data rescue by them in the case of crisis.

Public Announcements

Is the repository able to provide advanced warning to the scientific community and public about possible disruptions in service?

Notices can be posted on the website.

Are there peer organizations the repository can rely on in a crisis? Are there metadata and

Citizen Archival

Is the repository able to facilitate citizen archivists in coordinating and archiving the repository? Can the repository find out what has been archived by citizen archivists?

There is no metadata available to the public identifying datasets that are not known to be replicated elsewhere. There are minimal lines of communication with the public.

instructions on what the public should archive in case of crisis?

Section Themes Total Weighted Score

Single Repository Dataset Replication (Min=5, Max=24)	Repository Infrastructure (Min=4, Max=18)	Routine Operations (Min=3, Max=9)	Software (Min=6, Max=21)
0	0	0	0
Cybersecurity (Min=3, Max=9)	Sustainability (Min=6, Max=30)	Network (Min=4, Max=18)	
0	0	0	

is risks to their repository. A higher score is better.

Repository Name:

Repository Parent Organization:

Date:

Project:

Scenarios

More
resiliency

Fill in this
column only!

(Yes/No question
to be asked, "No" =
0)

Class 3

User Entered
Score. (Integers
only) Score
Range:
1 -Class 1
2 -Class 2
3 -Class
3

Suggested
Weight
(If score > 1)

Weighted Score

Scenario Weighted
Score (Zeros are
assigned to
categories that
would be impacted
during a crisis)

1: Natural
Disaster hits in
less than 48
hours

2: Repository
defunding by
parent
organization or
primary funder,
ex. government

ts. Datasets held by another organization in the same project (ICARUS at NSF NCAR vs

Multiple copies on multiple physical devices		1	0	Could any logical copies survive the crisis?	0	0
Three or more types of storage medium		1	0	Would any types of used storage medium survive the crisis?	0	0
Two or more offline copies		2	0	Could a secure offline copy be preserved and used after the crisis?	0	0
Three or more data facilities, geographically separated		2	0	Would a different data facility be able to maintain preserving and serving data during the crisis?	0	0
Two or more international data facilities		2	0	Would a different data facility be able to maintain preserving and serving data during the crisis?	0	0
Section Total Raw Score. (Min=5, Max=15)	0	Section Total Weighted Score. (Min=5, Max=24)	0	Section Total Weighted Scenario Score. (Min=0, Max=24)	0	0

There are multiple staff members for each role and staff in each role can cover for staff in other roles.		2	0	Are staff able to perform their duties?	0	0
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Data volume is small enough to effectively transfer all data within one day.		1	0	Is it possible to virtually move data in the time remaining?	0	0
There is capability to break down and physically move datasets in one hour notice.		1	0	Is it possible to physically move data in the time remaining?	0	0
There is redundant power/HVAC/emergency power generation for 24 hours or more.		2	0	Will the physical plant infrastructure keep cyberinfrastructure running?	0	0
Section Total Raw Score. (Min=4, Max=12)	0	Section Total Weighted Score. (Min=4, Max=18)	0	Section Total Weighted Scenario Score. (Min=3, Max=18)	0	0

Backups occur regularly and to one or more devices. Backups are tested on a regular schedule.		1	0	Can backups be restored from after the crisis?	0	0
SLAs are defined and publicly accessible.		1	0	Can SLAs and their deadlines be found by the public and followed during or after the crisis?	0	0
Required training for staff and time/funds for staff improvement via optional conferences, trainings, or certifications.		1	0	Can training continue after the crisis?	0	0
Section Total Raw Score. (Min=3, Max=9)	0	Section Total Weighted Score. (Min=3, Max=9)	0	Section Total Weighted Scenario Score. (Min=0, Max=9)	0	0

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The dataset can be made available as text (e.g. PDF file or CSV or just a plain printout or even photograph).

The software is packaged by the developer and possibly held in software repository indices. The repository has made images and backups of their current software and configuration, and can redeploy with minimal downtime.

The software is open source and has an active community supporting it.

Source code is well documented and easy to install. There are images available for deployment by non-technical users.

The software is in a common language and does not require niche software libraries and licensed compilers to execute. Code discipline is good and there is thorough test coverage of the software.

1

0

Is the primary software or alternatives available during the crisis?

0

0

1

0

Is the technical expertise to package the software systems the repository runs on available?

0

0

2

0

Is the technical expertise creating the software still available?

0

0

1

0

Can the software be installed by someone with no experience in an acceptable timeline?

0

0

1

0

Can the software be maintained by someone with no experience in an acceptable time frame?

0

0

User Interface and User Experience of the software is polished for all users. Documentation is clear and thorough. Sane defaults and examples are available for the software. Programmatic access is well documented and tested.

1

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Can the software be used by someone with no experience in an acceptable time frame?

0

0

Section Total Raw Score.
(Min=6, Max=18)

0

Section Total Weighted Score.
(Min=6, Max=21)

0

Section Total Weighted Scenario Score.
(Min=1, Max=21)

0

0

There is real time logging with alerts at the operating systems level and repository systems level. Alerts will notify staff of anomalous changes such as increased deletions or access. Logging is available for all facilities and are monitored before an update of off-site copies.

1

0

Is logging still working and being monitored during the crisis to ensure normal operations are occurring?

0

0

There are some accounts with higher permissions for modifying datasets and systems. There is multi factor authentication required for authentication and executing critical tasks like deletion of data. Logs around stopping of systems and deletion of deposits are reviewed periodically.

1

0

Are permissions being maintained and authorized access and actions being controlled to ensure normal operations during the crisis?

0

0

There are cybersecurity professionals available through a larger organization and a cybersecurity professional on staff.							
		1	0	Would cybersecurity staff be able to mitigate issues of the crisis?	0	0	
Section Total Raw Score. (Min=3, Max=9)	0	Section Total Weighted Score. (Min=3, Max=9)	0	Section Total Weighted Scenario Score. (Min=0, Max=9)	0	0	

Datasets/data systems are regularly modernized with links to tools for accessing them. There is input from the community on how modernization should be done.							
		2	0	Can data modernization or migration happen after the crisis?	0	0	
There are plans for a systematic shutdown of systems, transfer of assets, and care for staff.		1	0	Can repository shutdown plans be enacted during the scenario?	0	0	
There is a signed formal agreement with another data repository, with documented plans as to how to execute the transition.		2	0	Have the continuity agreement and plans been executed before or possibly during the scenario?	0	0	
There is a signed formal agreement with another data repository, with documented plans as to how to execute the transition.		2	0	Have the continuity agreement and plans been executed before or possibly during the scenario?	0	0	
There is a clear plan of what would happen to the repository in case of larger institutional crisis, including merging with another repository or moving to another unit in the institution.		1	0	Can the parent institution execute continuity plans during the crisis?	0	0	

There are multiple, roughly equivalent funding streams such that losing one or two is not a critical blow to operations.		2	0	Will funding streams continue during and after the crisis?	0	0
Section Total Raw Score. (Min=6, Max=18)	0	Section Total Weighted Score. (Min=6, Max=30)	0	Section Total Weighted Scenario Score. (Min=3, Max=30)	0	0

The repository is a member of one or more professional networks with reciprocal data sharing agreements or MOUs to serve as peer data copies.		2	0	Does a professional network replicate the holdings of the repository during the crisis?	0	0
There are other libraries, archives, or repositories in the larger organization that have plans to support archival in case of crisis.		1	0	Does a sibling repository replicate the holdings of the repository during the crisis?	0	0
Notices can be posted on the website, to professional societies, and across social media.		1	0	Can a notice be made to the public?	0	0
There are simple methods for identifying datasets the repository is the primary resource for. It is easy to bulk download full datasets in standard, machine readable formats. Community forums exist for the repository. There are methods for the repository to know what has been backed up by citizen archivists.		2	0	Are citizen archivists able to replicate data in the time available?	0	0

Section Total Raw Score. (Min=4, Max=12)	0	Section Total Weighted Score. (Min=4, Max=18)	0	Section Total Weighted Scenario Score. (Min=2, Max=18)	0	0
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Total Raw Score. (Min=31, Max=93)	0	Total Weighted Score. (Min=31, Max=129)	0	Section Total Weighted Scenario Score.	0	0
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scores are split in thirds by total weighted max		
Rubric Total Weighted Score < 43	43 <= Rubric Total Weighted Score <= 86	86 < Rubric Total Weighted Score
Low resiliency	Moderate resiliency	High resiliency

The minimum and maximum

(Scenario 1
Min=26,
Max=114)
(Scenario 2
Min=19,
Max=81)

3: Current
cyberattack,
cybercrime, or
other loss of
control over
cyberinfrastructure

4: Loss of
technical
expertise critical
to running
repository

UC Davis) must fill this out

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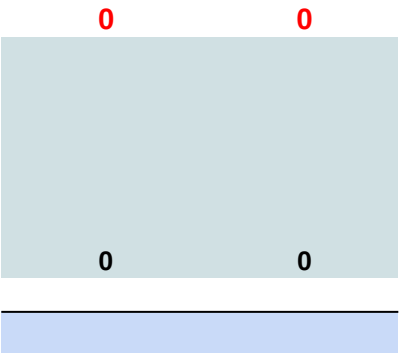
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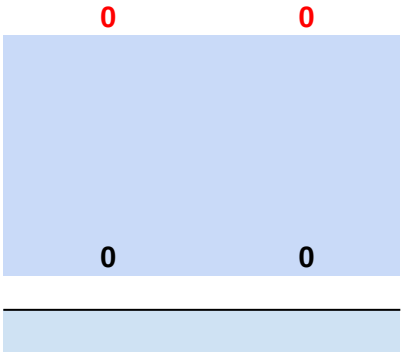


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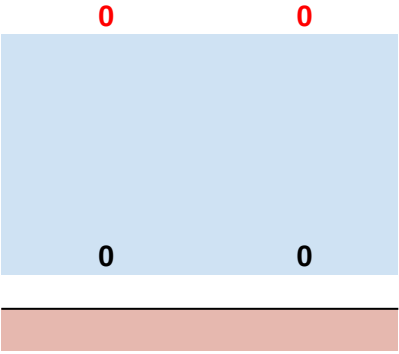
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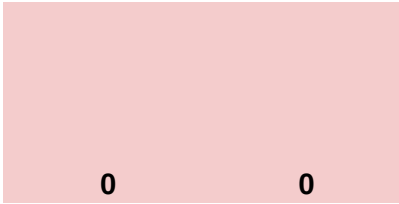
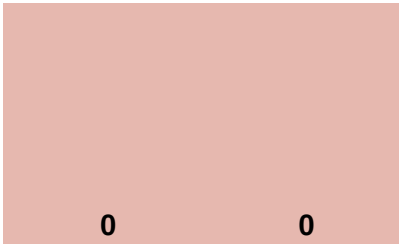
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possible scores for scenarios are:

(Scenario 3
Min=22,
Max=102)

(Scenario 4
Min=16,
Max=78)

The Data Impact and Recovery Scorecard (DIRS) is used to determine metadata resiliency
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Repository Crisis Scorecards Instructions

Repository Descriptors			Resiliency Metric	
Big Picture Question	Descriptor	Descriptor definition	Class 1	Class 2
Section Theme: Data and Metadata				
<u>Can the data be reconstructed if the central servers are offline? Look at https://metadata-handbook.org/</u>				
straightforward for others to understand and reuse your data and metadata?	Data Management Plans	describes all related publications and datasets?	attached to the dataset.	the places where the dataset is described.
	PID Coverage and Landing Pages Metadata Completeness	each resource have its own landing page? data/metadata? e.g., OAIS reference model completeness vocabularies, metadata schemas, ontologies, etc.?	landing pages or sitemap required and optional fields. FAIR is not considered as CARE principles evaluated for deprecation.	landing pages or sitemaps but not both. some have metadata for optional fields. Dataset considers FAIR, but is only human readable. implemented, such as Local Context notices. no formal process exists for evaluating dataset deprecation.
	FAIR Compliance	dataset by and for the people the data comes from?		
	Ethical Usage Considerations Data Deprecation Plans	for evaluating and deprecating datasets if needed?		
Section Theme: Impact				
repository access impact others relying on dataset access?	Support through Transition	monitoring, data pushed to the WMO for forecast models, etc? hurricanes, tornadoes, air quality, etc?	transformed to users. actions, i.e. issue similar warnings. and loss of access causes a hole. access to the deposits.	not be free and may not be public. a paid service for similar actions, i.e. pay for warnings. no plans to provide a partial solution. repository causes a partial loss of access to the deposits.
	Audience Impact	in networks relying on data hosted by members?		
	Federated Networks Replicating Other Repositories	go online, would there be no access to the datasets?		

and societal impacts. A higher score is better/more imp Repository Name:
Repository Parent Organization:
Date: Project:

resiliency	column only!	Suggested	Weighted Score
Class 3	0 -N/A (Impact only)	Weight	
	1 -	1 -	
health-ui.vercel.app/ for a summary of repository metadata			
publications and datasets.		1	0
a sitemap including all resources.		2	0
for required fields and optional fields.		2	0
machine actionable.		1	0
they are not applicable.		1	0
community as to appropriateness.		1	0
Max=18)	0	Max=24)	0

a different endpoint.		2	0
actions for tree with similar reach.		2	0
synced for changes.		1	0
access to the deposits.		1	0
Max=12)	0	Max=18)	0

[illegible]